

# Arthur Gautier

AI & Machine Learning Engineer | GenAI, NLP & Machine Learning | M.Sc. Data Science

Singapore | +65 8011 4754 | artgau79@gmail.com | linkedin.com/in/agtier | Portfolio

## Work Experience

### Data Scientist — SCOR

Singapore & Paris, France | Oct. 2024 – Present

- Designed and optimized OCR-to-LLM pipelines for insurance use cases, extracting structured data from P&C contracts and Life medical reports of up to 1,000 pages, targeting 90% accuracy.
- Contributed to AI Assistant, SCOR's GenAI platform used daily by 150+ underwriting and claims experts, driving up to 30% productivity gains on ~1M document pages/month.
- Built a parallelized Python pipeline for LLM evaluation across 115+ fields and 200+ documents, generating reference datasets from expert-validated data and automating prediction matching, semantic error detection, and human review.
- Developed an AI-powered benchmarking pipeline for the Sustainability team, extracting, normalizing, and comparing 72 criteria across annual reports from 10+ companies, cutting time from 1–2 weeks of manual work to 15 minutes.
- Built a production-grade Python package (Cox Proportional Hazards + XGBoost) for mortality-risk prediction from wearable IoT data, supporting SCOR's Biological Age Model.

### Data Scientist — Covéa (internship)

Niort, France | Jan. – Jul. 2024

- Designed an ETL pipeline and an open-source alternative (YOLO on Azure) to a managed Custom Vision service, on a 72M-image / 4M-customer-file automation project.
- Achieved 97% concordance with the production system on ~100k images while cutting costs by 64%, applying a FinOps-driven approach.

### AI Research Assistant — Bangkok University (internship)

Bangkok, Thailand | May – Jul. 2023

- Built a YOLOv8 computer vision system for automatic food recognition and pricing from meal-tray photos, deployed on a Raspberry Pi prototype with integrated camera and UI.

## Education

### M.Sc. Data Science & Processing, Information Systems — SeaTech, School of Engineering Toulon, France | 2021 – 2024

- Coursework: Image Processing, Signal Processing, Computer Vision, Numerical Methods for Differential Equations.

### Exchange Semester — Bergen University

Bergen, Norway | Aug. – Dec. 2023

- Coursework: Machine Learning (Random Forest, SVM, PCA, KNN), Information Theory, Concurrent Programming.

### Preparatory Class — Lycée Descartes

Tours, France | 2019 – 2021

- MPSI-MP — Intensive mathematics and physics curriculum with preparation for highly competitive entrance exams.

## Projects

**HanMap** (ongoing) — iOS app combining a semantic map of Mandarin vocabulary with a spaced-repetition (Anki) system for HSK learning.

**JobAgent** (ongoing) — Automated job application platform scraping job descriptions from URLs to generate tailored CVs and cover letters with LLMs, using user-created templates to preserve authentic candidate experiences.

**Chessboard photo converter** — Developed an end-to-end computer vision pipeline to detect a physical chessboard, recognize pieces, reconstruct the game state, and export the position to Lichess for analysis.

## Skills

**GenAI / NLP:** OCR-to-LLM Pipelines, RAG & Retrieval Strategies, Document Intelligence, Prompt Engineering

**ML & Stats:** Deep Learning (TensorFlow, Keras), Computer Vision (YOLO, OpenCV), Survival Analysis (Cox, XGBoost)

**Programming & Data Engineering:** Python, PySpark, Pandas, SQL

**Cloud, MLOps & DevOps:** Azure, Foundry, GCP, Databricks, Docker, FastAPI, CI/CD, Git/GitLab

**Languages:** English (Fluent), French (Native), Mandarin Chinese (Learning)

**Hobbies:** Running — 400m/800m competitions, marathon (Strava) · Cinema & Music (SensCritique)